

KAVISHKA HERATH

Final Year Computer Science Undergraduate – Colombo, Sri Lanka

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PROFESSIONAL SUMMARY

- Final year Computer Science undergraduate with foundational knowledge in Cloud Computing and DevOps. Experienced with AWS services, Docker, CI/CD pipelines, and Infrastructure as Code through hands on projects. Actively expanding expertise in modern DevOps tools, with a strong interest in automation, cloud security and designing scalable and efficient systems.

CORE DEVOPS & CLOUD SKILLS

- Cloud & Infrastructure:** AWS Core Services (EC2, S3, RDS, VPC, IAM, CloudFront, ACM, WAF), AWS ECS (Fargate), AWS Lambda, AWS Cost Monitoring & Optimization (Fundamentals)
- Infrastructure as Code (IaC):** Terraform, AWS CloudFormation (Fundamentals)
- Containers & CI/CD:** Docker, GitHub Actions, Secure CI/CD Pipelines, HTTPS Deployments, Auto-scaling & High Availability Architectures (Fundamentals)
- Monitoring & Observability:** AWS CloudWatch, Prometheus, Grafana (Metrics, Dashboards, Alerts Fundamentals)
- Security & Networking:** AWS Security Best Practices (IAM, Least Privilege, WAF), AWS Secrets Manager, Networking Fundamentals (VPC, Subnets, Routing, Security Groups)
- DNS & SSL Management:** Cloudflare (DNS Management, SSL/TLS Configuration)
- Systems & Automation:** Linux Fundamentals, Python Scripting & Automation, Git & Version Control
- Databases & Platforms:** PostgreSQL, MySQL, Kubernetes (Fundamentals)
- IDEs & Developer Tools:** Visual Studio Code, IntelliJ IDEA, Cursor

WORK HISTORY

Process Associate

Infomate (Pvt) Ltd – John Keells Holdings | Colombo, Western Province, Sri Lanka

Jul 2022 - Jun 2023 | Full-time | Hybrid

- Processed vendor invoices, ensuring accuracy and timely payments.
- Handled multilingual purchase orders and invoices (Swedish, Norwegian, Spanish, Finnish).
- Resolved discrepancies by liaising with internal teams and vendors, maintaining accuracy under tight deadlines.

MC Associate – YouTube Project

Orfium | Sri Lanka

Feb 2025 - Jul 2025 | Full-time | Hybrid

- Managed YouTube copyright claims for global music clients.
- Ensured content compliance with YouTube and partner guidelines.
- Conducted metadata review and conflict resolution.

Research Associate – TikTok & Meta Project

Orfium | Sri Lanka

Nov 2023 - Jan 2025 | Full-time | Hybrid

- Led content classification projects for TikTok, reviewing and categorizing multimedia content based on community guidelines.
- Contributed to model training data for AI-based moderation tools.

EDUCATION

CINEC Campus (Pvt) Ltd – Sri Lanka
BSc (Hons) Computer Science (Software Engineering) (University of Wolverhampton)
Final Year Undergraduate | 2023 - Present

CERTIFICATIONS

- Introduction to Image Generation – Google Cloud Skills Boost (Issued May 2025)
 - Introduction to Software Engineering – IBM (Issued Apr 2025)
 - Machine Learning & FPGAs – IEEE CINEC Student Branch (Issued Apr 2025)
 - Workflow Specialist – Asana (Issued Mar 2025)
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PROJECTS

SolarCast.AI - Full Stack AI Application

Feb 2025 - Nov 2025

- SolarCast.AI is a full stack AI/ML application for real time solar energy optimization by forecasting cloud coverage and solar irradiance. Built with a FastAPI backend, React/TypeScript frontend, PostgreSQL database, and integrated Google Earth Engine and OpenWeatherMap APIs, the platform leverages multiple ML models for cloud detection, forecasting, and energy output prediction. Features include interactive dashboards, real time analytics, model performance monitoring, JWT based authentication, and offline ready data handling. This project demonstrates end to end capabilities in AI driven renewable energy prediction, full stack development, and advanced geospatial analytics.

Sri Lankan Immunization Real Time Tracker

Mar 2024 - Jan 2025

- The Sri Lankan Vaccine Safety Network (SLVSN) is a web-based Big Data visualization dashboard to track adverse events from vaccination programs across Sri Lanka. Built using Laravel 11.31, PHP 8.2+, and MySQL, the system follows an Agile methodology and uses Change Data Capture (CDC) for real-time updates. It includes vaccine tracking, adverse event monitoring, and data analytics integrated with Power BI. The system enhances vaccine safety surveillance and ensures timely responses to safety concerns, supporting the reliability of Sri Lanka's vaccination programs.
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PUBLICATIONS

AI-Driven Cloud Cover Prediction to Enhance Solar Energy Efficiency

(06th CINEC International Research Symposium - 2025)

- I am currently leading an ongoing individual project that uses AI and satellite imagery to predict cloud cover and improve solar energy production efficiency. The project leverages advanced machine learning techniques such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) models for accurate cloud classification and forecasting. By integrating geographic information systems (GIS) with AI models, the goal is to optimize solar power generation and enhance solar panel efficiency. This project is still in progress, aiming to provide actionable insights that can improve solar energy performance through real-time cloud cover prediction.

From Data to Decisions: How Medical Data Helps Sri Lankan Medical Professionals To Manage Vaccine Adverse Event

(Blog article published on Medium platform)

- Developed a proof of concept web based dashboard, "Sri Lankan Vaccine Safety Network", to enhance vaccine safety monitoring across Sri Lanka. Leveraging primary survey data from healthcare professionals and anonymized secondary datasets, I designed a Laravel-based web application integrated with Power BI visualizations to track, analyze, and report post-vaccination adverse events. The system features interactive dashboards displaying temporal trends, demographic analyses, geographic distributions, and vaccine comparisons, with offline data entry support for rural areas. Built with strong data protection measures aligned with Sri Lanka's Personal Data Protection Act, the platform ensures secure access, encryption, audit logging, and consent management.
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REFERENCES

Mr. Gihan Perera
Ph.D. Student in Biochemistry
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Dr. Ananda Jayawardana
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